Fiscal Year:	FY 2009	**Task Last Updated:**	FY 01/30/2009
PI Name:	Fornace, Albert M.D.		
Project Title:	Mouse models approach for intestinal tumorigenesis estimates by space radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	http://www9.georgetown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Start/end dates changed to 4/1/07-3/31/11, per update from P. Denkins/JSC; previously had 12/1/2006-11/30/2010 (9/2007)		
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Task Description:	Ionizing radiation (IR) is a known risk factor for colorectal (CR) cancer. Considering the high spontaneous incidence of gastrointestinal (GI) cancer in the US and even higher incidence of pre-malignant lesions, such as colonic polyps, an even modest increase by radiation exposure could have a significant effect on health risk estimates for manned space flight. During space travel, astronauts are exposed to levels of radiation, which increases their risk for acute short-term deleterious effects, such as during a solar storm, and for long term deleterious effects like cancer. Long duration space missions, like those at the International Space Station, lunar missions, or a mission to Mars, are of sufficient duration to allow for appreciable cumulative doses of cosmic radiation that could have profound health consequences. This project is important because there are no good estimates for intestinal, particularly CR, cancer by space radiation. Considering the increased CR cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors and to investigate modulation of specific signaling pathways as a model for cancer prevention. We are employing an animal model, APC+/min, that has been successfully used to demonstrate radiation-induced intestinal cancer. The APC+/min model has inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. The strategy is to compare results for space radiation, protons and HZE (56Fe), on intestinal tumor development to that of low-LET (linear energy transfer) radiation (gamma rays). Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for humans exposed to space radiation. Many of the molecular events leading to intestinal tumors have been characterized. We will monitor and compare molecular changes in three types of radiation-induced tumors. The specific aims for this proposal are: 1. Quantitatively compare HZE ion or high-energy proton irradiation to gamma-ray effects on intestinal tumor incidence and grade using the Apc+/min tumor model. 2. Characterize and compare inactivation of wt APC locus on chromosome 18B1 after gamma vs. HZE ion or high-energy proton irradiation with the Apc+/min tumor model. 3. Determine intestinal tumor incidence in the Apc+/min; Ppm1d-/- mouse model and mechanisms for reactivation of the Apc+/min tumor phenotype by radiation. 4. Determine molecular signatures by comparative genomic hybridization (CGH) and expression profiling in spontaneous intestinal tumors compared to those induced by HZE ions, high-energy protons, and gamma irradiation. 5. Quantitatively compare HZE ion or high-energy proton irradiation vs. gamma-ray effects on incidence of other tumors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the incidence of CR cancer in 2002 was 53.6 per 105 with a death rate of 20.1 per 105 or about 57,000 deaths per year. Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and CR carcinogenesis.
Task Progress:	The overall plan is to apply the well-characterized APC +/min mouse model of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, the relative toxicity of 56Fe particles and protons to gamma -rays needed to be determined so that we can standardize with doses equitoxic to 2 and 5 Gy of gamma -radiation in C57Bl/6J mice; these gamma -ray doses have been shown to be effective in increasing intestinal tumorigenesis. The rationale for this initial study is that acute toxicity for the planned radiation sources in mice is not available. We have carried out four runs so far at the NSRL at Brookhaven National Laboratory. Results to date indicate that the relative biological effectiveness (RBE) for 56Fe particles compared to gamma -rays is 1.25 and RBE for proton is 1.06. Although the RBE for protons was expected, the RBE for 56Fe particles is surprisingly low and is an important parameter that will allow us to plan long-term experiments. Irradiated mice are being observed long-term, and specific tissues are also being provided to various NASA investigators to complement their own projects. We also irradiated 6 to 8 weeks old female APC+/min mice with iron particles and protons with 5 Gy equivalent of gamma -radiation. Including gamma-radiation a total of 82 mice was irradiated and have been monitored for intestinal tumorigenesis. Female mice were used to also monitor for mammary tumorigenesis which has been reported to be increased by IR. Mice were observed for 100 days and at the end of the observation period mice are being humanely euthanized as per Georgetown University Animal Care and Use Committee (GUACUC) approved protocol. The APC+/min mice show frequent spontaneous tumors and IR typically triggers a modest GI tumor increase in adults compared to larger increases in newborn or younger mice, and we also found a modest but significant increase after gamma -rays. In contrast, the increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater. Interestingly, in the course of our analyses, we also noted a larger increase of clustered tumors with space radiation compared to gamma -rays. Results to date indicate the space radiation, both high-energy protons and 56Fe particles, may have unique properties conferring an increased tumorigenic risk compared to gamma rays.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Datta K, Doiron K, Fornace A Jr. "Low- and high-LET radiation induced chronic stress responses in mice: Implications for carcinogenesis." Annual meeting of the American Association for Cancer Research, San Diego, CA, April 12- 16, 2008. Annual meeting of the American Association for Cancer Research, San Diego, CA, April 12- 16, 2008. , Apr-2008

Abstracts for Journals and Proceedings	Datta K, Hyduke D, Doiron K, Fornace A Jr. "High-LET radiation induces greater chronic oxidative stress but lesser transcriptomic alteration than low-LET radiation: Implications for carcinogenesis." Annual meeting of the Radiation Research Society, Boston, MA, September 21- 24, 2008. Annual meeting of the Radiation Research Society, Boston, MA, September 21- 24, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Datta K, Hyduke D, Doiron K, Fornace A Jr. "Low- and high-LET radiation induced chronic stress and intestinal tumorigenesis: a mouse model study." NASA Space Radiation Investigators' meeting, Philadelphia, PA, June 29 to July 2, 2008. NASA Space Radiation Investigators' meeting, Philadelphia, PA, June 29 to July 2, 2008. , Jul-2008